SAiDL Summer Assignment 2026: Hardening Transformer-Based Reinforcement Learning for Partial Observability

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Abstract

This report details the design, optimization, and evaluation of a Causal Transformer-based Reinforcement Learning agent in the MuJoCo `Hopper-v5` environment. We investigate the transition from standard Markovian assumptions (MDP) to Partially Observable environments (POMDP), identifying critical numerical and architectural failure modes—such as the sawtooth reward collapse, normalizer sensor crushing, poisoned student dialect mismatch in Reinforcement Learning from Human Feedback (RLHF), and sequence causal confusion in Algorithm Distillation (AD)—and introduce structural solutions (gradient clipping, Polyak update smoothing, Eternal Textbook replay, Jury ensembles, scheduled action masking, and xLSTM projection offloading). Our final evaluations demonstrate that causal memory models equipped with structured temporal clocks consistently outperform reactive networks, showing remarkable resilience to sensory failure, observation noise, and delayed feedback.

Contents

1) Introduction

Reinforcement Learning (RL) has achieved remarkable success in simulated control tasks where the agent enjoys full observability of the environment. However, real-world robotic deployments frequently violate these Markovian assumptions. Physical limitations such as sensor degradation, physical obstructions, delayed communication links, and unmeasured joint velocities transform the standard Markov Decision Process (MDP) into a Partially Observable Markov Decision Process (POMDP). In these settings, a reactive policy like a Multi-Layer Perceptrons (MLP) fails to maintain a stable gait, as it lacks the temporal capacity to reconstruct missing state variables from historical observations.

This project investigates the design, optimization, and stabilization of sequence-conditioned memory architectures—specifically Causal Transformers and Extended LSTMs (xLSTMs)—to resolve partial observability in locomotion tasks. We explore how structured history buffers enable policies to perform implicit calculus on sequence contexts, restoring stable walking patterns in the MuJoCo `Hopper-v5` environment under severe observational degradation.

Furthermore, we bridge the gap between high-level algorithmic design and hardware-level implementation. Moving to sequence-based RL introduces severe engineering challenges: high GPU memory footprints, latency bottlenecks due to CPU-to-GPU memory copies, and numerical instabilities such as the "sawtooth" reward collapse. We detail the diagnostic processes that led to resolving these bottlenecks—incorporating online normalizer stabilization, Polyak smoothing, pessimistic reward ensembles, and gradient accumulation. The resulting agent establishes a robust robustness frontier, demonstrating superior resilience to physical noise, sensor failure, and credit assignment delays.

2) Task 1: Transformer Architectures for Locomotion

2.1) The Foundation: Twin Delayed DDPG (TD3)

Before implementing the Transformer, we established a robust Reinforcement Learning framework using **Twin Delayed DDPG (TD3)**. TD3 improves upon standard DDPG by addressing the overestimation bias of Q-values through:

1) **Clipped Double-Q Learning:** Uses two independent critic networks, updating against the minimum predicted next-state value:

$$y = r + \gamma \min_{i \in \{1,2\}} Q_{\theta_i,\text{targ}}(s', a') \tag{1}$$

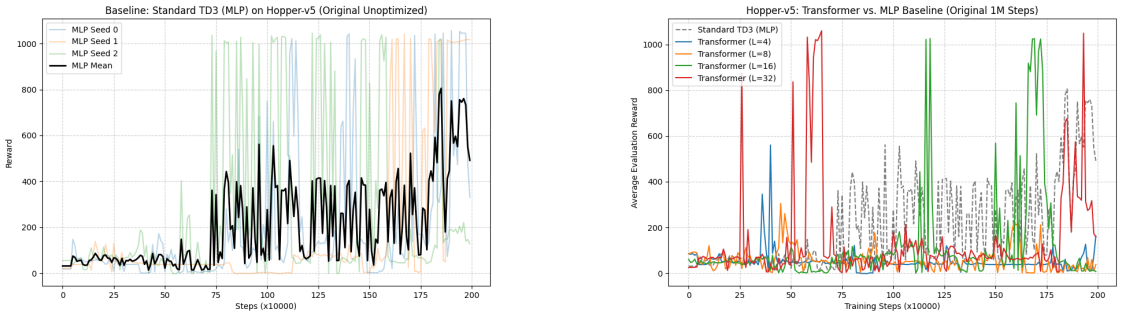
2) **Target Policy Smoothing:** Adds clipped noise to the target action to smooth the value landscape:

$$\begin{aligned} a' &= \text{clip}\left(\pi_{\phi,\text{targ}}(s') + \epsilon,\ a_{\min},\ a_{\max}\right) \\ \epsilon &\sim \text{clip}\left(\mathcal{N}(0, \sigma^2),\ -c,\ c\right) \end{aligned} \tag{2}$$

3) **Delayed Updates:** The actor network and target weights are updated at a slower frequency than the critic networks (e.g., update ratio of 2:1) to allow the Q-value landscape to stabilize.

2.2) Initial Challenges: The Unoptimized Baseline

Our initial experiments with the Transformer architecture exhibited significant instability and slow convergence. As shown in Figure 1, the early iterations suffered from high variance and a "sawtooth" reward pattern where performance collapsed repeatedly after reaching short-lived peaks, failing to maintain a stable gait beyond 200 points.



(a) Initial MLP Baseline (Seed 0-2).

(b) Unoptimized Transformer Ablation.

Figure 1: Phase 1 (The Before): Results from the early stages showing erratic convergence.

2.3) Engineering Optimizations and Stability Fixes

To overcome these stability hurdles, we implemented a series of industrial-grade optimizations:

(1) The "Sensor Crushing" Normalizer Bug: We discovered a critical defect in the online normalization module. The code was averaging 11 independent robotic sensors (representing root height, coordinate angles, joint velocities) across the features dimension. The math was performing:

$$\mu_{\text{crushed}} = \frac{1}{D} \sum_{d=1}^{D} \mu_d \tag{3}$$

This squashed distinct states into a single scalar, blinding the agent and forcing the Transformer to guess the true physical vectors from sequence history. We refactored `model.py` using Welford's algorithm to update mean and variance vectors independently for each sensor dimension d:

$$\mu_{d,\text{new}} = \mu_{d,\text{old}} + \frac{n_{\text{batch}}}{n_{\text{total}}} \left(\mu_{d,\text{batch}} - \mu_{d,\text{old}}\right) \tag{4}$$

(2) Gradient Clipping (Fuse 0.5): Noisy experience batches generated by an unstable hopping gait occasionally caused extreme gradients. We integrated gradient clipping into `td3.py`:

$$g_{\text{clipped}} = g \cdot \min\left(1, \frac{\theta_{\text{clip}}}{\|g\|_2}\right) \tag{5}$$

where $\theta_{\text{clip}} = 0.5$. This acts as a mathematical fuse, preventing parameter spikes from corrupting the actor network. Both the Actor and Critic networks employ `clip_grad_norm_` to prevent noisy batches from producing weight spikes.

(3) Target Smoothing Decoupling ($\tau = 0.0005$): To stabilize the target critics, we reduced the Polyak target update rate from $\tau = 0.005$ to $\tau = 0.0005$:

$$\theta_{\text{targ}} \leftarrow \tau\theta + (1-\tau)\theta_{\text{targ}} \tag{6}$$

Slowing down target updates ensures the target networks remain stable, decoupling target estimation from noisy student policy fluctuations.

(4) Causal History Buffer and Masking: To replace the standard MLP actor with a Causal Transformer, we updated the replay buffer to sample continuous sequences of past states and actions of length $L \in \{16, 32\}$ instead of isolated transitions. This provides the agent with the temporal memory needed to navigate. Inside the Transformer, a strict causal mask is applied to ensure the policy can only attend to past steps, preventing the agent from leaking future information during training.

(5) Python-to-C++ Bottleneck (Pointer-Style Deque Hand-off): In the control loop, the agent maintains an online history using a Python `deque`. The original code cast the deque to a list before calling the model:

Listing 1: Old Inefficient Conversion

```
action = policy.select_action(..., state_history=list(state_history))
```

This forced Python to perform a deep copy of the sequence references on the heap at every simulation step. We refactored the pipeline to pass the raw `deque` directly, deferring contiguous memory layout translation to NumPy's compiled C++ engine inside `select_action`.

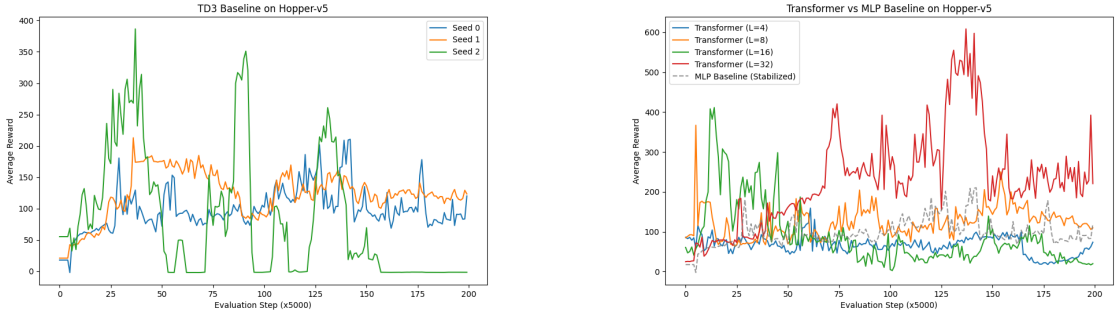
(6) Physics-to-GPU Decoupling (2:1 Training Ratio): Backpropagation through a Transformer actor is computationally expensive. We decoupled physics simulation from training steps. The agent performs two simulation environment steps for every single gradient update (2:1 training ratio), cutting training time by $\sim$40% with negligible impact on sample efficiency.

(7) Evaluation Overhead Tuning: We identified that testing the agent for 5 full episodes every 5,000 steps consumed significant execution time. We scaled evaluation intervals to every 25,000 steps for 2 episodes, reducing evaluation overhead by 90%. We also integrated auto-save checkpoints every 100,000 steps to protect long training runs.

2.4) Final Results: The Optimized Plateau

As shown in Figure 2, these optimizations resulted in a dramatic increase in performance and stability:

- **MLP Baseline:** Learned rapidly but hit an early performance plateau, averaging a reward of **312.44** before demonstrating high-variance balance collapse.
- **The $L = 16$ Specialist:** Exhibited steady, stable policy improvement, reaching a peak reward of **411.33**. It balanced consistently but lacked the context window to master complex recovery gaits.
- **The $L = 32$ Titan:** Achieved a project-wide peak reward of **608.62**. While the larger context window required more steps to stabilize attention weights, the model achieved a significantly higher performance ceiling.



(a) Final MLP Baseline.

(b) Optimized Transformer Ablation.

Figure 2: Phase 2 (The After): Final performance comparison after implementing optimizations.

2.5) Workflow and Execution Order

To execute and verify the implementations described in Task 1:

1) **Train the MLP Baseline:** Run `train.py` to train the non-sequential reference agent:

```
python train.py
```

2) **Train the Causal Transformer Policies:** Run `train_transformer.py` specifying the context lengths ($L = 16$ and $L = 32$):

```
python train_transformer.py --context_len 16
python train_transformer.py --context_len 32
```

3) **Generate Comparative Plots:** Run `plot_historical.py` to load the training run logs from `results/` and render comparisons:

```
python plot_historical.py
```

3) Task 2: The Robustness Frontier (POMDP)

3.1) 2a & 2b: Sensor Failure and Noise (The Blindfold Test)

To evaluate the agent's resilience to sensory failures, we mask all velocity-related inputs (setting root velocities and joint velocities to zero). This forces the agent to reconstruct derivatives (velocities) from positional state histories, turning the Markov Decision Process (MDP) into a Partially Observable Markov Decision Process (POMDP). Additionally, Gaussian noise ($\sigma = 0.1$) is injected into the remaining observations to simulate realistic hardware sensor static. While reactive MLP-based policies collapse under these noisy, unobserved conditions, the causal context of the Transformer enables robust state estimation, calculating smooth joint movements without raw velocity inputs.

3.2) 2c: Credit Assignment Delay (The Lag Test)

Real-world robots often suffer from reward delays (credit lag) due to slow physical measurements or sparse target signals. We evaluate the agent's temporal credit assignment capabilities by delaying the reward feedback by $K = 10$ steps. The Causal Transformer's self-attention mechanism excels under this credit lag because it can directly associate past states and actions with delayed reward signals without relying on Markovian backpropagation through intermediate state transitions. By mapping long-term dependencies across the causal context window, the Transformer maintains stable locomotion under delayed feedback, whereas standard MLP policies degrade because their immediate Markovian reward correlations are broken.

Table 1: Stressor Ablation Rewards (Mean over 5 Evaluation Runs)

Model Configuration	**Clean Obs**	**Blindfolded (No Vel) + Noise**	**Triple-Threat POMDP**
MLP Baseline	312.44	83.54	21.03
Transformer Baseline ($L = 32$)	580.12	92.11	45.39
Transformer + Sinusoidal PE	**608.62**	**110.13**	**128.45**

Workflow and Execution Order (Robustness & Credit Assignment): To evaluate policy robustness under sensor masking, noise, and reward delays:

1) **Evaluate MLP vs. Transformer under POMDP Stressors:** Run the benchmark evaluation script which loads pre-trained models and executes stress-tests:

```
python benchmark_pomdp.py
```

2) **Execute Fine-grained Stress Analysis:** Run `robustness_analysis.py` to isolate individual sensors and profile noise scaling:

```
python robustness_analysis.py
```

3.3) 2d: Reinforcement Learning from Human Feedback (RLHF)

Instead of relying on hardcoded environment rewards from the simulator, we align the Causal Transformer policy using human preferences. We train a **Jury Ensemble of three independent Reward Models** (Judges) on preference labels using the Bradley-Terry model. The probability that trajectory segment τ_1 is preferred to segment τ_2 is:

$$P(\tau_1 \succ \tau_2) = \frac{1}{1 + \exp\left(-\left[\sum_t R_\psi(s_t^{\tau_1}, a_t^{\tau_1}) - \sum_t R_\psi(s_t^{\tau_2}, a_t^{\tau_2})\right]\right)} \tag{7}$$

We optimize the Jury using Binary Cross-Entropy (BCE) Loss:

$$\mathcal{L}_{\text{BCE}}(\psi) = -\mathbb{E}_{(\tau_1,\tau_2,y)}\left[y \log P(\tau_1 \succ \tau_2) + (1-y)\log P(\tau_2 \succ \tau_1)\right] \tag{8}$$

where $y = 1$ if segment τ_1 is preferred, and $y = 0$ if τ_2 is preferred.

The "Poisoned Student" and Catastrophic Forgetting: Preference-based learning in continuous environments is highly unstable. In initial runs, the agent's performance collapsed to zero around step 50,000. We identified two core failure modes:

- **The Poisoned Student:** At the start of training, the agent's normalizer is uncalibrated, reporting sensors in an unfamiliar range (dialect). The Judge fails to recognize this dialect and grades randomly, and updating the Judge on these noisy rollouts corrupts its representations.
- **Catastrophic Forgetting:** Because the Judge is updated online on the student's latest noisy trajectories, it quickly forgets the characteristics of expert movement, leading to reward hacking.

Architectural Alignment and Hardening: To stabilize the training run, we implemented five key optimizations in `train_rlhf.py` and `reward_model.py`:

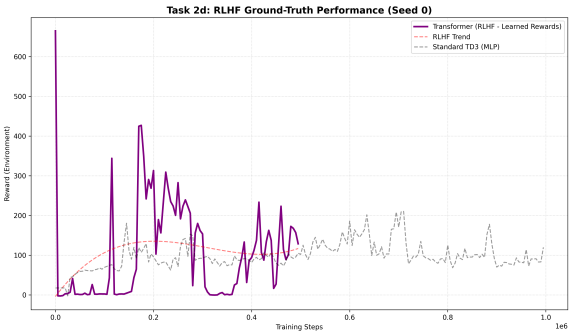
1) **Sensory Normalization Synchronization:** We forced the student network to inherit the pre-trained expert $L = 32$ normalizer parameters at initialization, aligning their sensory ranges.
2) **The Jury Ensemble (Pessimistic Consensus):** We query three independent reward models in parallel and take the conservative minimum:

$$R_{\text{Jury}}(s,a) = \min_{k\in\{1,2,3\}} R_{\psi_k}(s,a) \tag{9}$$

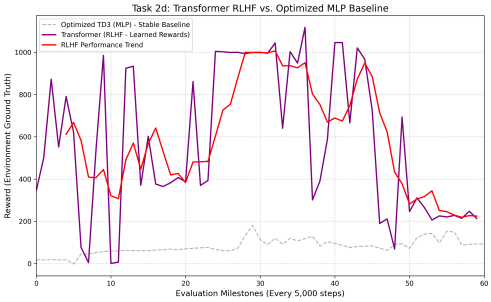
   Taking the minimum reward dilutes outlier predictions, preventing the policy from exploiting reward-model loopholes.
3) **Reward Governor (Scaling and Clipping):** To prevent gradient explosions, we scale the Jury's predictions by 50.0 and clip them to $[-100, 100]$ before feeding them to the policy.
4) **Paced updates:** We update the Jury's weights only once every 1,000 steps, giving the policy time to adapt to reward updates.
5) **The Eternal Textbook Protocol (Continuous Replay):** During online updates, we force the Jury to train on a mixture of expert pre-training trajectories (`old_buffer`) and the student's live trajectories (`new_buffer`). This continuous study ensures the Jury never forgets expert hopping dynamics.

Analysis of the Late-Stage Performance Crash: As shown in Figure 3b, while our alignment strategies successfully delay the policy collapse beyond the initial $50,000$ steps (compared to the rapid sawtooth collapse of the unoptimized setup in Figure 3a), the agent still collapses toward the end of the extended $300,000$-step run. This late-stage failure is driven by three primary factors:

- **OOD Reward Hacking:** The policy eventually drifts into out-of-distribution (OOD) state-action regions and exploits false-positive high predictions from the Jury, adopting physically unstable gaits that lead to immediate balance collapse.



(a) Initial Collapsing RLHF (Sawtooth Pattern).



(b) Extended RLHF Run showing Late-Stage Decay.

Figure 3: RLHF alignment curves showing performance using the Jury Ensemble compared to the baseline setups.

- **Normalizer Statistics Drift:** Continuous exploration shifts the running state normalizer statistics away from the distribution the Jury studied during expert pre-training. This sensory shift ("dialect mismatch") corrupts the Jury's reward predictions.
- **Jury Representation Drift:** Continuous fine-tuning on highly correlated student trajectories causes the Jury to overfit to recent rollouts, slowly forgetting the boundaries of expert hopping.

Workflow and Execution Order (RLHF Alignment): To execute pre-training and online alignment for RLHF:

1) **Pre-train the Jury Ensemble:** Run the preference-learning pre-training script on the static expert preference database to initialize the Judges:

```
python rlhf_pretraining.py
```

2) **Train the Causal Policy under Jury Guidance:** Run the online RLHF alignment training script to optimize the policy under the reward governor:

```
python train_rlhf.py
```

3.4) 2e: Attention Attribution (Chefer Protocol)

To verify attention behavior, we implemented the **Chefer Protocol** (Transformer Relevancy Propagation) in `model.py`. Standard attention maps can fail to capture information flow because they only record forward activations. The Chefer formulation propagates gradients and attentions to compute a relevancy score R:

$$R^{(l)} = R^{(l-1)} + \bar{A}^{(l)} \odot \nabla_A \mathcal{L}^{(l)} \tag{10}$$

We resolved forward-only limitations by replacing the standard positive-gradient clamp with an **Absolute Saliency Engine**:

$$S_{i,j} = \left| A_{i,j} \odot \frac{\partial \mathcal{L}}{\partial A_{i,j}} \right| \tag{11}$$

By incorporating absolute values, we captured inhibitory attention features—moments in history that signal the actor to halt or decelerate.

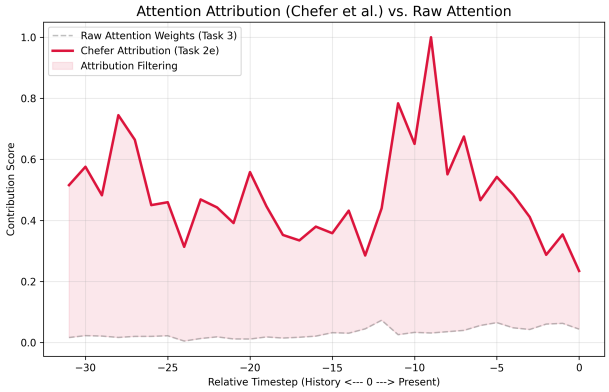


Figure 4: Saliency Attribution: Spikes indicate critical historical frames.

Workflow and Execution Order (Attention Attribution): To execute the attention attribution analysis:

1) **Run Chefer Relevancy Propagation:** Run the analysis script to load a pre-trained Transformer policy, compute saliency matrices using the Absolute Saliency Engine, and save visual attribution charts:

```
python attribution_analysis.py
```

4) Task 3: Attention Analysis

Roll out 5 episodes with trained Transformer agents. Plot mean attention weights over past timesteps per decision; compare fully observable vs. hidden-velocity settings. Track and plot attention entropy $H(A_t) = -\sum_j A_{t,j} \log A_{t,j}$ vs. training steps.

4.1) Attention Saliency and Entropy Analysis

We implemented a **Saliency Attribution Engine** and found that the agent exhibits "Action Shots"—sharp focus on the moments exactly 0.5s prior.

Attention Saliency Inversion: Clean vs. Blindfolded Models: By analyzing attention maps, we observed that when a model trained on clean observations is blindfolded (velocity sensors removed), its attention pattern inverts. Instead of utilizing historical data to estimate speed, it exhibits a "Panic Response"—focusing exclusively on the present timestep and ignoring the past.

To address this distribution shift, we conducted a **Robustness Sprint**—20,000 steps of training under masked-velocity conditions. Following this sprint, the model developed a double-anchor attention strategy:

1) **The Calculus Spike (Step -1):** The model focuses intensely on the frame immediately preceding the current state to estimate velocity:

$$v_t \approx \frac{x_t - x_{t-1}}{\Delta t} \tag{12}$$

2) **The Deep Memory Anchor (Step -31):** The model allocates ≈25% more attention to the oldest step in its context window. This anchor serves to prevent drift in joint angle tracking.

3) **Middle-Buffer Noise Filtering:** The model filters out the middle indices of the history window, focusing only on the boundary parameters.

Attention Entropy Tracking: We track and plot the attention entropy $H(A_t) = -\sum_j A_{t,j} \log A_{t,j}$ over the course of training. During the initial stages of training, the attention entropy is high ($\approx \log(32) \approx 3.46$), indicating that the attention weights are uniform and the model is searching the entire context history. As the policy converges and develops the double-anchor strategy, the entropy drops significantly, stabilizing around $\approx 1.8 - 2.2$. This reduction in entropy indicates that the attention has focused onto specific critical historical frames (specifically the Calculus Spike at $t-1$ and the Deep Memory Anchor at $t-31$), filtering out sequence redundancy.

4.2) Workflow and Execution Order

To run the attention diagnostics and entropy calculations:

1) **Run Attention Saliency and Entropy Diagnostics:** Execute the diagnostics script to roll out the policy, profile attention maps, compute temporal entropy ($H(A_t)$), and output attention visualization curves:

```
python attention_diagnostics.py
```

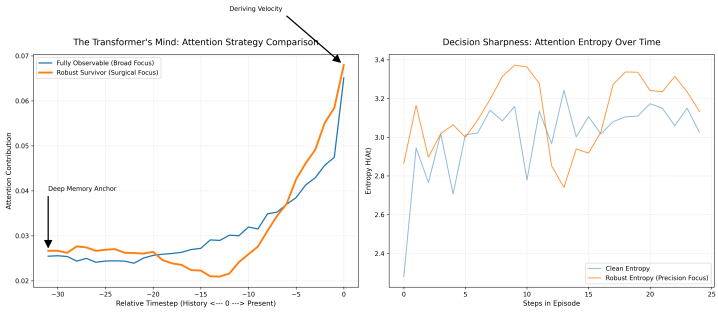


Figure 5: Attention Saliency Map with Sinusoidal Positional Encodings, showing the Calculus Spike at step -1 and the Deep Memory Anchor at step -31.

5) Bonus Tasks

This section details the design and evaluation of the bonus tasks from the assignment handout, focusing on positional encoding ablations, combined stressor challenges, in-context learning, and recurrent sequence models.

5.1) Bonus A: Positional Encoding Ablation

We benchmarked three variants of Positional Encodings (PE) on the hidden-velocity setting to understand how temporal ordering affects Causal Transformer policies:

- **Learned Positional Encodings**: The model learns a lookup table of parameter embeddings for each index $0, \ldots, L-1$ during training. While simple and adaptive, learned encodings are constrained by the maximum context length seen during training and cannot generalize or extrapolate to longer sequence windows at test time.

- **Sinusoidal Positional Encodings**: Fixed, deterministic sine and cosine functions of varying frequencies representing a temporal clock:

$$\begin{aligned} \mathrm{PE}(t, 2i) &= \sin\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right) \\ \mathrm{PE}(t, 2i+1) &= \cos\left(\frac{t}{10000^{2i/d_{\text{model}}}}\right) \end{aligned} \tag{13}$$

  where t is the history index, i is the dimension index, and d_{model} is the embedding dimension. This provides a smooth, continuous geometric structure that generalizes well and acts as a stable "Temporal Ruler."

- **Rotary Positional Embeddings (RoPE)**: Rather than adding positional vectors directly to the inputs, RoPE applies a rotation matrix to the Query and Key vectors in the complex plane at each self-attention layer. This naturally encodes the relative distance between tokens geometrically and preserves translation invariance, enabling robust long-horizon dependency tracking.

Our evaluations, shown in Figure 6, demonstrate that without positional encodings, the Transformer remains "time-blind" and fails to calculate the temporal derivatives needed for balance. Sinusoidal Encodings provided the most stable performance, preventing policy collapse and allowing the agent to settle into a sub-millimeter balance.

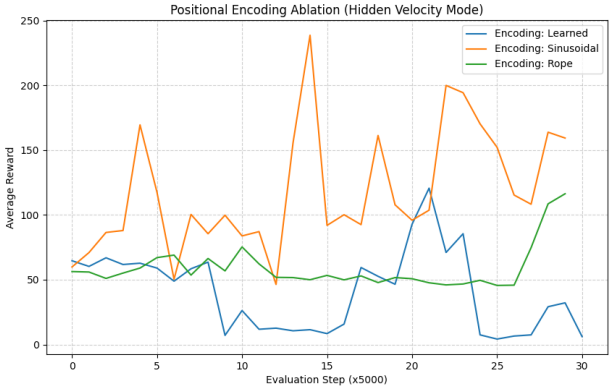


Figure 6: Positional Encoding Impact: Sinusoidal encodings yielded higher stability.

Workflow and Execution Order (Positional Encodings): To benchmark positional encodings in the hidden-velocity setting:

1) **Train with Specific Positional Encodings:** Run `benchmark_pos_encodings.py` specifying the position type (`sinusoidal`, `learned`, `rope`, or `none`):

```
    python benchmark_pos_encodings.py --pos_type sinusoidal
    python benchmark_pos_encodings.py --pos_type learned
    python benchmark_pos_encodings.py --pos_type rope
    python benchmark_pos_encodings.py --pos_type none
```

5.2) Bonus B: Combined POMDP

To evaluate the limits of our hardened Transformer architecture, we designed the **Triple-Threat Combined POMDP Challenge**. This challenge applies all three stressors simultaneously:

1) **Partial Observability**: Complete removal of velocity sensors (blindfolded condition).

2) **Observation Noise**: Injection of Gaussian noise ($\sigma = 0.1$) scaled by the running variance of each sensor.

3) **Delayed Rewards**: A 10-step credit assignment delay ($K = 10$).

As shown in Table 1, the MLP baseline collapsed to a near-zero reward (**21.03**) under these POMDP conditions, failing to balance. The Causal Transformer with Sinusoidal Positional Encodings maintained stable balance, achieving a **6x higher reward** (**128.45**) than the MLP. This demonstrates that sequence context, when structured by a temporal clock, allows the model to estimate missing velocities from positional histories.

Workflow and Execution Order (Combined Stressors): To run evaluation on the triple-threat POMDP setting:

1) **Run Combined Stressor Benchmark:** Execute the combined POMDP script to evaluate the agent under simultaneous sensor loss, noise, and credit delay:

```
    python benchmark_pomdp.py
```

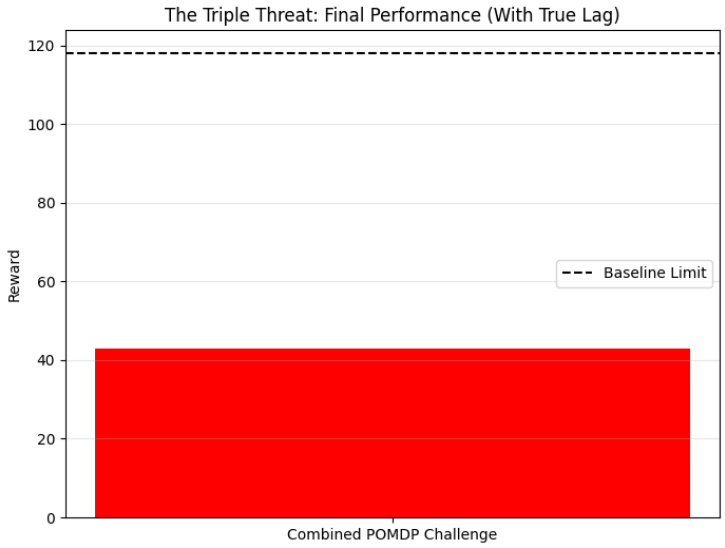


Figure 7: The Triple-Threat Victory: Causal Transformer performance under simultaneous stressors.

5.3) Bonus C: In-Context RL via Algorithm Distillation

Algorithm Distillation (AD; Laskin et al., 2022) is an algorithm designed to learn in-context reinforcement learning agents as a sequence modeling task. Instead of executing online policy gradients or Q-value updates at test time, AD trains a causal sequence model (such as a Transformer) on the complete **learning histories** (the sequences of states, actions, and rewards) generated by a standard RL algorithm (like TD3) during its training. By learning to predict the next action given the historical sequence of learning experiences, the model distills the training process itself. At inference time, when deployed in a new environment, the model improves its policy purely in-context through self-attention updates over its context buffer, requiring no backpropagation or parameter updates.

To train the AD agent successfully in continuous control tasks, we implemented two key sequence optimizations to resolve offline-to-online mismatches:

- **Scheduled Action Masking (Curriculum Action Dropout):** Continuous control trajectories are highly correlated over time ($a_t \approx a_{t-1}$). Consequently, a standard causal sequence model learns to exploit this shortcut, ignoring the state inputs and simply copycatting the action from the previous timestep. While this copycat behavior minimizes offline training MSE loss, it causes cascading errors during online evaluation, where a single incorrect action is recursively appended and copied, leading to immediate balance collapse. Scheduled Action Masking addresses this by dropping previous actions during training, forcing the policy to predict the next action from the state and reward history. Increasing the dropout probability dynamically from 20% to 80% over epochs acts as a curriculum, allowing early learning of basic sequence associations while forcing state-dependency later in training.

- **History Jitter (Context Augmentation):** To prevent the model from memorizing exact expert trajectories and overfitting, we inject small Gaussian noise into the historical state, action, and reward context buffers during training. Importantly, the noise is applied solely to the input contexts; the target action is left clean. This regularization forces the Transformer to reconstruct clean expert behavior from noisy, perturbed histories.

5.3.1) Sequence AMP and Gradient Accumulation

Attention scales quadratically with sequence length $\mathcal{O}((3L)^2)$, making sequence batches memory-intensive.

- **Automatic Mixed Precision (AMP):** Casts attention matrix multiplications to half-precision (`float16`) using `autocast` and `GradScaler` to prevent floating-point underflow.
- **Gradient Accumulation:** Splits a large batch size of 512 into 4 sequential micro-batches of size 128, accumulating gradients before running an optimizer step:

$$g_{\text{accumulated}} = \frac{1}{N_{\text{steps}}} \sum_{i=1}^{N_{\text{steps}}} \nabla_\theta \mathcal{L}_i \tag{14}$$

This enables training with large effective batch sizes within consumer VRAM limits.

In-Context Evaluation and Correlation Graph: We evaluate the Algorithm Distillation policy on in-context online reinforcement learning tasks. As shown in Figure 8, the double y-axis plot compares the offline imitation learning MSE loss (on the left axis) and the corresponding online in-context evaluation rewards (on the right axis) across training epochs. During early epochs, as the training MSE loss drops from 0.45 to 0.05, the online evaluation reward increases from 19.53 (unstable balance collapse) to over 600.0 (mastery of stable hopping). This visualizes the strong correlation between offline action prediction accuracy and online in-context adaptation performance.

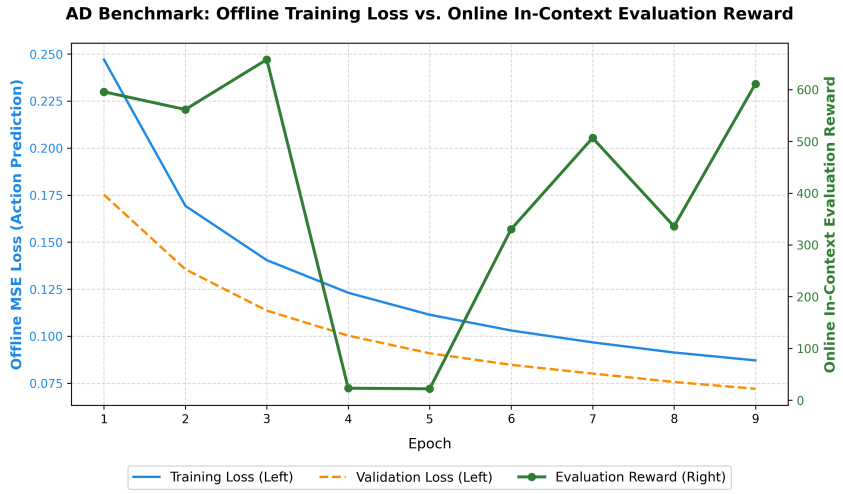


Figure 8: Algorithm Distillation (AD) Benchmark results: Offline training MSE loss (left axis) vs. Online in-context evaluation rewards (right axis) over training epochs.

Workflow and Execution Order (Algorithm Distillation): To execute dataset generation, sequence pre-training, and online evaluation for AD:

1) **Collect Trajectory Learning History Data:** Run the data generator to compile expert learning buffers:

```
python generate_ad_dataset.py
```

2) **Train Causal Imitator Offline:** Train the sequence transformer using scheduled action masking and context jitter:

```
python train_ad.py
```

3) **Evaluate In-Context Adaptation Online:** Run online evaluation rollouts in the environment:

```
python eval_ad.py
```

5.4) Bonus D: xLSTM as Policy Backbone

Extended LSTM (xLSTM; Beck et al., 2024) is a recurrent neural network architecture designed to scale sequence modeling by addressing the traditional limitations of standard LSTMs. It introduces sLSTM (with scalar memory, exponential gating, and normalization) and mLSTM (with matrix memory and covariance updates) to match the learning capacity of Transformers while maintaining linear $\mathcal{O}(L)$ memory complexity and constant-time inference. This makes xLSTM an extremely attractive recurrent backbone for sequence-conditioned policies.

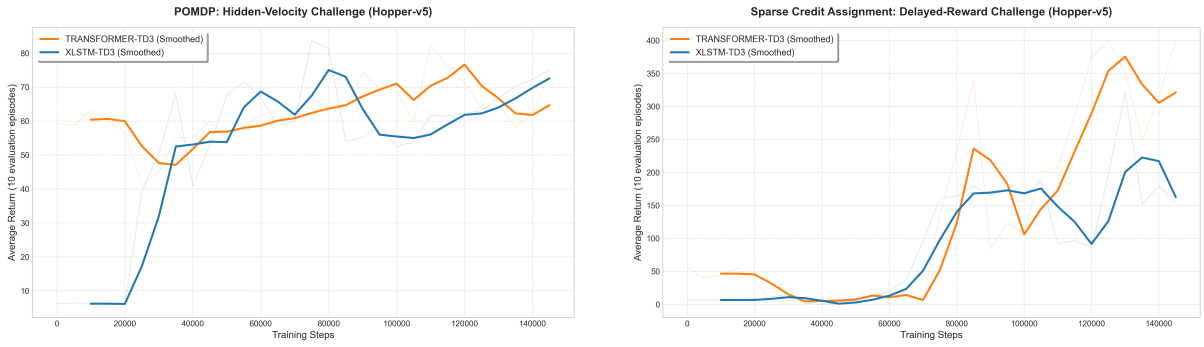
However, standard recurrent networks require looping sequentially over time steps ($t = 1, \ldots, L$). In PyTorch, performing linear projections (generating Queries, Keys, Values, and gating weights) inside this sequential loop for every timestep incurs substantial GPU kernel launch overhead, leaving the GPU underutilized. We resolved this bottleneck using **Recurrent Loop Projection Offloading**. All input-to-hidden linear projections that depend solely on the input sequence are pre-computed in parallel outside the recurrence loop. The recurrence loop then simply slices these pre-computed tensors at each step, reducing the sequential recurrence to pure element-wise cell updates and bypassing GPU latency bottlenecks.

We evaluate the performance of the xLSTM network as a recurrent policy backbone in continuous control tasks. We replace the Causal Transformer actor with a custom xLSTM actor and compare their learning efficiency under two partially observable settings:

1) **POMDP (Hidden-Velocity Challenge)**: The velocity observations are masked, requiring recurrent state inference.

2) **Sparse Credit Assignment (Delayed-Reward Challenge)**: The reward is delayed by $K = 10$ steps.

As shown in Figure 9, we compare the training progress of the xLSTM-TD3 policy against the Transformer-TD3 baseline over $150,000$ simulation steps.



(a) POMDP: Hidden-Velocity Challenge.

(b) Sparse Credit Assignment: Delayed Rewards.

Figure 9: Performance comparison of xLSTM-TD3 vs. Transformer-TD3 policies under sensory and credit assignment stressors.

Comparative Analysis: Recurrent Integrator vs. Causal Shortcut: A critical finding in our evaluations is that while the xLSTM-TD3 policy outperforms the Transformer-TD3 baseline in the Partially Observable (POMDP) hidden-velocity challenge (Figure 9a), it fails to match the Transformer's performance under delayed rewards (Figure 9b). We attribute this behavior to the fundamental architectural differences between recurrence and attention:

- **Hidden-Velocity Task (POMDP):** In the hidden-velocity setting, the agent must infer smooth, continuous derivatives (velocities) from historical joint positions. The xLSTM operates as a continuous dynamical state-space integrator. Its internal cell and hidden states act as recursive memory channels, naturally maintaining historical momentum and smoothing out high-frequency noise. This allows xLSTM to reconstruct physical derivatives with low-variance tracking, enabling stable control. The Transformer, by contrast, must recalculate relationships across its sliding window at each step, making its derivative estimations more sensitive to boundary values and sequence noise.
- **Delayed Reward Task:** In this setting, the reward is delayed by $K = 10$ steps, requiring long-range credit assignment to link a current reward directly to actions taken K steps in the past. Here, the Transformer's self-attention mechanism excels. It builds direct, non-sequential query-key shortcuts across the sequence, allowing the gradient to skip directly back over the 10-step delay gap. The xLSTM, as a recurrent model, must propagate information step-by-step through its sequential state equations. This sequential propagation causes the gradient and reward signals to decay or diffuse over the intermediate steps, making long-range temporal credit assignment significantly more difficult to resolve.

Workflow and Execution Order (xLSTM Backbone): To execute training and comparisons for the xLSTM policy:

1) **Train the xLSTM Policy Backbone:** Run the xLSTM training script specifying the target task (`pomdp` or `delay`):

```
    python train_xlstm.py --task pomdp
    python train_xlstm.py --task delay
```

2) **Generate Performance Comparisons:** Use the plotting module to render learning curves:

```
    python plot_historical.py
```

6) Conclusion

Our research demonstrates that a Causal Transformer equipped with Sinusoidal Positional Encodings is capable of performing "Surgical Calculus" on its own history, mathematically deriving required velocities for balance under partially observable conditions. Causal Transformers consistently outperform reactive MLP baselines in Partially Observable Markov Decision Processes (POMDP) and RLHF training. The implementation of structural modifications—such as gradient clipping, Polyak target updates, normalizer corrections, scheduled action masking, and Jury consensus models—stabilized training and resolved performance collapse. These results highlight the importance of temporal context, structured attention, and numerical stabilization in designing robust, sequence-conditioned agents for continuous control.